

Cross-Sectional Short-Horizon Equity Direction Prediction: A Walk-Forward Machine Learning Study on the S&P 500

Tyrese Rome Tan-Gatue

August 17, 2026

Abstract

This report covers a machine learning pipeline built to predict short-term (5-trading-day) relative stock price direction across a 55-stock, sector-balanced subset of the S&P 500, and to turn those predictions into a long-short quintile portfolio. The pipeline combines 28 technical indicators with 85 monthly commodity price series from the World Bank’s Commodity Markets Outlook, a purged walk-forward validation scheme, and a CatBoost classifier retrained periodically over a 2021–2026 sample. Pooled out-of-sample accuracy came in at 51.8%, against a 50.6% base rate, with a pooled AUC of 0.534. That points to a very weak but real edge over the study period, and one that visibly decayed in the more recent folds. Feature importance analysis shows the entire predictive signal is coming from technical and price-based features; the 85 commodity features contributed zero measurable importance despite doubling the feature space, and the honest conclusion is that this study did not turn up a strong, persistent signal.

Contents

1	Objective	2
2	Universe Construction	2
2.1	Stock Selection	2
2.2	Sample Period	2
3	Feature Engineering	3
3.1	Technical / Price-Based Features (28 features)	3
3.2	Commodity Features (85 features)	3
3.3	Normalization	3
4	Label Construction	4
5	Model and Validation Design	4
5.1	Classifier	4
5.2	Walk-Forward, Purged, Periodic-Retrain Scheme	4

6	Results	5
6.1	Pooled Out-of-Sample Performance	5
6.2	Performance by Year	5
6.3	Long-Short Portfolio	5
6.4	Feature Importance	6
7	Discussion and Limitations	6
8	Conclusion	7
9	Acknowledgments	7

1 Objective

My goal of this was to build and evaluate an end-to-end pipeline for predicting short-term stock price direction and to use those predictions to construct and paper-trade a diversified portfolio. The original design in mind was to use live/historical data via the Alpaca API; due to access issues, price history was instead sourced from **yfinance** and cached locally as a Parquet file, with commodity data supplied separately via the World Bank’s Commodity Markets Outlook (CMO) monthly release.

2 Universe Construction

2.1 Stock Selection

The investable universe was built by:

1. Scraping the current S&P 500 constituent list (ticker, GICS sector) from Wikipedia.
2. Computing each stock’s Average Dollar Volume (ADV), defined as

$$\text{ADV}_i = \frac{1}{T} \sum_{t=1}^T \text{Close}_{i,t} \cdot \text{Volume}_{i,t}, \quad (1)$$

used as a liquidity proxy because it normalizes traded volume by price level, making liquidity comparable across low- and high-priced names.

3. Selecting the top 5 stocks by ADV within each of the 11 GICS sectors, yielding a fixed, sector-balanced universe of $5 \times 11 = 55$ tickers.

This design choice trades away some diversification benefit (only 5 names per sector) in exchange for a tractable, liquid, and sector-neutral cross-section suitable for a relative-ranking (long-short) strategy.

2.2 Sample Period

Daily OHLCV data was pulled from **yfinance** for 2021-01-01 through 2026-07-01, giving $55 \text{ stocks} \times \text{approximately } 1,378 \text{ trading days} \approx 75,790 \text{ stock-days}$ of raw data.

3 Feature Engineering

Feature engineering was the largest component of the notebook by volume of code and, as the results in Section 6 show, the component that determined whether any signal existed at all. Two feature families were constructed.

3.1 Technical / Price-Based Features (28 features)

Computed independently per ticker using `pandas_ta`, then combined:

- **Momentum/oscillators:** RSI at 5, 14, and 30-day windows; MACD; Bollinger Band %B; ADX(14).
- **Volatility regime:** realized volatility at 5, 20, and 60-day windows, plus the ratios σ_5/σ_{20} and σ_{20}/σ_{60} to detect volatility regime shifts.
- **Returns:** 1, 5, 10, 21, and 60-day price returns.
- **Trend regime:** price relative to 50/100/200-day moving averages; a trend-strength ratio $|\text{return}_5/\sigma_{20}|$; and 20-day rolling stability of 5-day momentum.
- **Volume:** 20-day volume z-score; On-Balance Volume (OBV) and its 20-day z-score.
- **Regime-shift detection:** rolling 20-day correlation between 5-day returns and 20-day volatility.
- **Cross-sectional features** (computed per date, across the 55-stock universe): relative strength (z-scored 5-day return vs. universe), market breadth (% of names with positive 5-day return), volatility percentile rank, and momentum dispersion (cross-sectional std. of 5-day returns).

3.2 Commodity Features (85 features)

Monthly commodity prices from the World Bank CMO dataset (metals, energy, agriculturals, fertilizers, etc.) were merged onto the daily equity panel using `pd.merge_asof` with backward direction. Two precautions were taken:

1. A 1-month publication lag was applied to every commodity series before merging, reflecting the fact that a given month’s commodity print is not published until roughly the following month.
2. Series missing $\geq 50\%$ of history (Barley, Sorghum) were dropped, leaving 85 usable series.

Because commodity values are identical across all 55 stocks on a given date, a naive rolling z-score computed on the row-sorted panel would incorrectly treat 55 duplicated rows as 55 independent observations. This was fixed by computing the rolling mean/std of each commodity on the unique daily series and broadcasting the z-score back to all stock-rows sharing that date. It’s a subtle bug to catch, but an important one: without the fix, the rolling window would span roughly $60/55 \approx 1$ trading day of true calendar time instead of 60.

3.3 Normalization

All 113 features (28 technical + 85 commodity) were cross-sectionally z-scored within each date, i.e.

$$z_{i,t}^{(f)} = \frac{x_{i,t}^{(f)} - \mu_t^{(f)}}{\sigma_t^{(f)}}, \quad (2)$$

where $\mu_t^{(f)}$ and $\sigma_t^{(f)}$ are the cross-sectional mean and standard deviation of feature f across the universe on date t . This is appropriate for a ranking/classification target defined relative to the

cross-section (Section 4), since it puts every stock’s feature value in the context of its peers on the same day rather than its own historical range.

4 Label Construction

The prediction target is a binary, cross-sectionally relative 5-day forward return rank:

$$\text{fwd_ret}_{i,t} = \frac{\text{Close}_{i,t+5} - \text{Close}_{i,t}}{\text{Close}_{i,t}} - 1, \quad (3)$$

$$\text{label}_{i,t} = \begin{cases} 1, & \text{if } \text{rank_pct}_t(\text{fwd_ret}_{i,t}) > 0.5, \\ 0, & \text{if } \text{rank_pct}_t(\text{fwd_ret}_{i,t}) \leq 0.5. \end{cases} \quad (4)$$

i.e. a stock is labeled 1 if its 5-day forward return is in the top half of the 55-stock universe on that date, 0 otherwise. Framing the problem this way (relative rank rather than absolute sign of return) removes market-wide drift and turns the task into a genuine stock-picking / relative-strength problem, which is the correct target for a long-short strategy.

5 Model and Validation Design

5.1 Classifier

I decided to use CatBoost Classifier as our model and I asked AI to configure it for speed and regularization on a fairly small per-fold training set, and it came up with the parameters: depth 4, 80 iterations with early stopping (patience 20), L_2 leaf regularization 5, 80% row subsampling, 70% column sampling per level, and a minimum of 20 samples per leaf.

5.2 Walk-Forward, Purged, Periodic-Retrain Scheme

Rather than a single train/test split or a naive k -fold CV (which would leak future information into the past for time-series data), I use a walk-forward scheme:

- **Training window:** rolling 500 trading days.
- **Retrain frequency:** every 20 trading days (roughly monthly), producing 57 populated folds (out of 68 possible windows) covering January 2022 through June 2026.
- **Purge gap:** because the label at date t uses information through $t + 5$, training dates within 5 calendar days of the test window start are excluded (**purge_cutoff**), preventing the classic time-series leakage where a training label’s forward window overlaps the test period.
- **Early-stopping validation split:** within each fold, 20% of the training data is held out (stratified) for early stopping, separate from the true out-of-sample test fold.
- **Minimum breadth filter:** dates with fewer than 10 stocks with valid data are dropped, and folds where either the training or test set is single-class are skipped (none were, in this run).

This is a reasonable and reasonably rigorous design for a cross-sectional equity signal: it avoids the two most common leakage failure modes in financial ML (shuffled CV and unpurged adjacent windows), and the periodic retrain lets the model adapt to changing regimes rather than relying on one static fit for five years.

6 Results

6.1 Pooled Out-of-Sample Performance

Across all 57 folds (pooled 1,100 predictions per fold, except the final partial fold of 440):

Metric	Model	Naive Baseline
Pooled accuracy	0.518	0.506
Pooled AUC	0.534	0.500

The model beats the majority-class baseline by roughly 1.2 percentage points of accuracy and achieves an AUC 3.4 points above random. That’s a real, measurable edge, but an economically thin one, and it’s in line with the broader literature on how hard short-horizon equity returns are to predict out of sample.

6.2 Performance by Year

Year	Accuracy	AUC	% Folds Beating Baseline
2021	0.570	0.612	100.0%
2022	0.507	0.519	41.7%
2023	0.529	0.542	76.9%
2024	0.521	0.542	75.0%
2025	0.521	0.533	69.2%
2026 (through June)	0.487	0.508	16.7%

Two things stand out here:

1. **Signal decay.** 2021 shows the strongest AUC (0.612), which makes sense given it’s effectively the warm-up period where training data is scarcest and closest to the training window itself. Performance decays toward, and eventually below, the 2022–2025 plateau of roughly 0.52–0.54 AUC by 2026, where only 1 of 6 folds beat baseline. That pattern is consistent with either genuine alpha decay (the market adapting to or arbitraging away the pattern) or with the feature set’s edge being concentrated in the unusual volatility regime of 2021–2022 and not generalizing well to later, calmer or differently structured markets.
2. **The final fold looks like an outlier and shouldn’t be taken at face value.** Fold 56 (2026-06-18 to 2026-06-30) reports accuracy of 0.384 against a local baseline of 0.809, with an AUC of 0.433. That 0.809 baseline is itself a red flag. It implies the test set’s label distribution was extremely imbalanced, roughly 81/19, on a target that’s constructed to be close to 50/50 by design through the cross-sectional rank. This is almost certainly an artifact of the short, partial final window (only 440 predictions, likely a truncated last month with an unusual number of missing forward-return dates near the end of the sample) rather than a real collapse in the model’s edge, and it should be treated as a data-boundary effect rather than folded uncritically into the year-level average.

6.3 Long-Short Portfolio

Constructing a long-short portfolio from the top vs. bottom predicted quintile each period:

Statistic	Value
Mean daily spread	0.135%
Std. daily spread	3.308%
Daily Sharpe ratio	0.041
Annualized Sharpe ratio	0.650
Fraction of positive days	52.4%

An annualized Sharpe ratio of 0.65 is directionally positive but well below the threshold (roughly 1.0 to 1.5) generally considered a viable systematic strategy before costs, and *no transaction costs, borrow costs, market-impact, or slippage are modeled anywhere in this notebook*. Given a 55-name universe traded via a rank-based long-short quintile at roughly monthly rebalance, realistic costs would eat into the already-thin 0.135% mean daily spread, and quite plausibly wipe it out entirely.

6.4 Feature Importance

CatBoost’s built-in feature importance on the final fold’s model attributes **100% of total importance to the 28 technical/price features and 0.00% to all 85 commodity features**, without exception. The top-ranked features were:

Feature	Importance (%)
vol_60d (60-day realized volatility)	17.3
adx_14 (trend strength)	10.6
macd	8.9
price_vs_ma_200	7.6
return_10d	7.0
rsi_30	4.9

The dominance of longer-window volatility and trend features (vol_60d, ADX, MA-200 deviation) over short-window oscillators suggests the model is picking up on *regime* (is this stock in a high-vol / trending state) more than on short-term mean-reversion or momentum *timing* signals.

7 Discussion and Limitations

1. **The core finding is a null-to-weak result.** A pooled AUC of 0.534 and an accuracy edge of 1.2 points over baseline is not a strong, tradeable signal. It is a small, real deviation from random that is consistent with (but does not prove) the existence of weak, decaying cross-sectional predictability in short-horizon technical patterns.
2. **Commodity data added complexity without adding signal.** Doubling the feature space from 28 to 113 features contributed a measured 0% of model importance. That’s a useful negative result. It suggests either that macro commodity levels have no genuine short-horizon linkage to individual-stock 5-day relative returns at monthly granularity, or that the frequency mismatch between 85 slow-moving, 2-month-lagged monthly series and a 5-day-horizon daily target simply makes the signal too coarse for the tree model to use. Either way, the feature engineering effort here would have been better spent on additional technical or cross-sectional constructs than on macro data at this frequency.

3. **2021 performance likely overstates steady-state edge.** With only a 500-day rolling window and a 2021 start date, the earliest folds are trained on the smallest, most idiosyncratic samples (dominated by the unusual 2021 reopening-trade / meme-stock volatility regime). The fact that AUC is highest exactly in this period and decays afterward is more consistent with regime-specific overfitting than with a stable, persistent edge.
4. **No cost model.** The reported 0.65 annualized Sharpe is gross of transaction costs, bid-ask spread, market impact, shorting costs/availability, and any slippage from executing a 55-name, rank-based rebalance. Given the small magnitude of the mean daily spread (0.135%), even conservative cost assumptions (a few bps per name per rebalance) could turn this into a flat or negative net Sharpe.
5. **Single train/model lineage.** All folds share one train window length (500 days), one retrain cadence (20 days), and one hyperparameter set, chosen once rather than tuned or validated against alternative configurations. The reported performance is therefore the result of a single design point in a much larger space of walk-forward configurations, and its robustness to that choice is untested.

8 Conclusion

This project produced a pipeline of a liquidity- and sector-aware universe, a purged and periodically retrained walk-forward validation scheme free of the most common time-series leakage bugs, and a cross-sectionally well-posed prediction target. Despite that and despite substantial feature engineering effort spanning 113 technical and macro features, the resulting signal is weak. Pooled AUC lands at 0.534, the accuracy edge over baseline is roughly one percentage point, and the gross annualized long-short Sharpe of 0.65 would need to survive real trading costs to mean anything economically. The most useful technical takeaway is that *all* of the model’s discriminative power traced back to price and technical features, with zero contribution from a large, carefully lagged commodity dataset. That’s a reminder that feature relevance, not feature quantity, is what determines whether a machine learning pipeline finds a signal at all. On the evidence gathered here, this pipeline did not identify a strong or clearly persistent trading signal.

9 Acknowledgments

I would like to acknowledge Claude (Anthropic) and DeepSeek for their assistance during the development of this project. These tools were used to accelerate code development, troubleshoot implementation issues, clarify technical concepts, and explore alternative approaches during the research process. Their assistance allowed me to iterate more efficiently on the implementation and test the underlying hypothesis.